CyGRC Platform — Module 2 (Cyber Defense)

Final Sprint Plan (Complete Operations MVP)

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1. Executive Overview & Scope

Module 2 builds the Cyber Defense Operations arm of CyGRC across 4 disciplines: SOC, VAPT, DFIR, and Threat Intelligence. It leverages the live **Shared Platform Foundation** (Keycloak Auth, PostgreSQL RLS, FastAPI Gateway, Next.js UI container) built during M1 Sprints 1 & 2.

Core MVP Scope Included:

- **SOC SIEM & Incidents:** Wazuh alert queue, Sigma rule converter (pySigma), NIST SP 800-61 Incident Case Management.
- **VAPT Scanner:** Asset discovery inventory, NVD & CISA KEV CVE lookup, CVSS prioritization formula, Jira integration.
- **DFIR Forensics:** Velociraptor collection API, append-only Chain of Custody log (chain_of_custody table).
- **Threat Intelligence & Red Team:** MISP/OpenCTI STIX 2.1 & TAXII 2.1 feed ingestion, Atomic Red Team execution, MITRE ATT&CK; mapping.
- **Cyber AI SOC Copilot:** AI alert triage and threat analysis with mandatory human approval guardrails.

2. Functional Team Structure (Alphabetical)

Team	Focus Area	Primary Responsibilities
Team A	SOC SIEM & Alert Ingestion	Wazuh SIEM integration, Sigma rule parser (pySigma), Alert Queue UI, NIST SP 800-61 Case Management.
Team B	VAPT Scanner & CVE Engine	Asset discovery inventory, NVD & CISA KEV CVE lookup, CVSS prioritization formula, Jira ticket push.
Team C	DFIR Forensics & Red Team	Velociraptor collection, append-only Chain of Custody log (chain_of_custody), Atomic Red Team execution harness.
Team D	Threat Intel & Cyber AI Copilot	MISP/OpenCTI STIX 2.1 & TAXII 2.1 feed ingestion, AI SOC Copilot with human approval guardrails.

3. Detailed Day-by-Day Work Execution Plan

Day 1 – Base Setup & Ingestion Endpoints

- **Team A (SOC SIEM):** Setup /api/v1/alerts API endpoints & Alert Queue UI skeleton.
- **Team B (VAPT):** Create Asset Inventory database table (assets) & API (/api/v1/assets).
- **Team C (DFIR & Red Team):** Create append-only chain_of_custody database table (REVOKE UPDATE/DELETE).
- **Team D (Threat Intel & AI):** Create Threat Intel STIX 2.1 objects store table & schema.

Day 2 – Core Ingestion Engines & Collectors

- **Team A (SOC SIEM):** Build Wazuh SIEM alert listener & Sigma rule parser (pySigma) converting Sigma YAML to Wazuh rules.
- **Team B (VAPT):** Build Asset Discovery pipeline (nmap / cloud inventory ingestion) with tenant scoping.
- **Team C (DFIR & Red Team):** Implement Velociraptor endpoint collection API (POST /api/v1/forensics/collect).
- **Team D (Threat Intel & AI):** Build TAXII 2.1 Server ingestion endpoints (/taxii2/) for external intel feeds.

Day 3 – Incident Cases & Threat Feeds

- **Team A (SOC SIEM):** Build NIST SP 800-61 Incident Case Management API (GET/POST /api/v1/cases).
- **Team B (VAPT):** Ingest NVD & CISA KEV daily CVE feeds (CVSS v3.1 scores & active exploitation flags).
- **Team C (DFIR & Red Team):** Build Chain of Custody logging helper & API to track artefact transfers and custodian signatures.
- **Team D (Threat Intel & AI):** Implement MISP (via PyMISP) and OpenCTI GraphQL client integrations.

Day 4 – Triage UI & CVE Prioritization

- **Team A (SOC SIEM):** Build Incident Case Management UI allowing SOC analysts to triage alerts into cases.
- **Team B (VAPT):** Implement Vulnerability Prioritization Formula (Priority = CVSS + KEV_bonus + criticality - mitigating_control).
- **Team C (DFIR & Red Team):** Implement Atomic Red Team execution harness for single-technique tests.
- **Team D (Threat Intel & AI):** Build STIX 2.1 automatic threat intel enrichment for incoming SOC alerts.

Day 5 – Real-Time Telemetry & ATT&CK; Mapping

- **Team A (SOC SIEM):** Connect Wazuh alert queue with real-time UI updates (Server-Sent Events / polling).
- **Team B (VAPT):** Build Vulnerability Findings UI (/vapt/findings) displaying prioritized CVEs.
- **Team C (DFIR & Red Team):** Build MITRE ATT&CK; v19.1 technique mapping matrix (15 Tactics, 222 Techniques).
- **Team D (Threat Intel & AI):** Build AI SOC Copilot service (POST /api/v1/ai/draft) suggesting triage questions and case notes.

Day 6 – Playbooks & Ticket Integration

- **Team A (SOC SIEM):** Implement IR workflow state machine in Case Management UI with state transition buttons.
- **Team B (VAPT):** Implement Jira API integration to automatically push remediation tickets for high CVE findings.
- **Team C (DFIR & Red Team):** Build Red Team Campaign Management API (/api/v1/campaigns) and ATT&CK; coverage heat map.
- **Team D (Threat Intel & AI):** Enforce Cyber AI Guardrails (zero auto-action, draft watermark is_draft=true, human review token).

Day 7 – Single-Click Case Handoff & GRC Signal Feed

- **Team A (SOC SIEM):** Build single-click Case-to-Forensics Handoff: Escalating a SOC case auto-triggers Velociraptor collection & opens DFIR case.
- **Team B (VAPT):** Connect VAPT scan results to feed Module 1's Compliance Engine as Automated Evidence.
- **Team C (DFIR & Red Team):** Build DFIR Incident Investigation UI with timeline view and evidence downloader.
- **Team D (Threat Intel & AI):** Build AI Threat Analyzer to generate structured incident summaries upon case closure.

Day 8 – High-Severity Approvals & Automated Scanning

- **Team A (SOC SIEM):** Build SOC Lead high-severity incident sign-off enforcement.
- **Team B (VAPT):** Add automated scan scheduling for network & web application assets.
- **Team C (DFIR & Red Team):** Build DFIR Chain of Custody audit export view.
- **Team D (Threat Intel & AI):** Connect Module 2 SIEM alert signals to feed Module 1's Control Status Engine as automated evidence.

Day 9 – Load Testing & System Validation

- **Team A (SOC SIEM):** Perform high-volume alert ingestion load test on Wazuh queue and verify sub-5s latency.
- **Team B (VAPT):** Verify Jira ticket sync and CVE priority calculations with test datasets.
- **Team C (DFIR & Red Team):** Execute Chain of Custody tamper test: Verify SHA-256 hash verification blocks altered files.
- **Team D (Threat Intel & AI):** Run end-to-end test of STIX/TAXII threat intel enrichment and M2-to-M1 evidence feeding.

Day 10 – Final Dry-Run & Client Handover

- **Team A (SOC SIEM):** Final SOC SIEM health check and alert queue verification.
- **Team B (VAPT):** Final VAPT scanner pipeline & Jira integration check.
- **Team C (DFIR & Red Team):** Final DFIR Chain of Custody audit verification and ATT&CK; matrix check.
- **Team D (Threat Intel & AI):** Finalize documentation package, API guide, and submit Module 2 to CyArt Tech LLP.